

COMP90077

Lecture 9

Linear Programming

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Caution

- Slides are designed as presentation aid to go along with lecture, not as study material
- Slides not designed to make sense on their own

Today

- Last time: Use LPs as black box for approximation algorithms
- Linear Programming Theory
- Duality

Recall: LPs

Linear Program

n variables, m constraints

Coefficients: $[a_1, \dots, a_n] = a$

$$\begin{bmatrix} b_{1,1} & b_{1,2} & b_{1,3} & \dots & b_{1,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{m,1} & b_{m,2} & \dots & \dots & b_{m,n} \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$$

Canonical form

$$\max \sum_i a_i \cdot x_i$$

$$\text{s.t. } b_{1,1}x_1 + \dots + b_{1,n}x_n \leq c_1$$

$$\vdots$$

$$b_{m,1}x_1 + \dots + b_{m,n}x_n \leq c_m$$

$$x \geq 0$$

Solving LPs – Geometric View

Example LP

$$\max \quad x_1 + 6x_2$$

s.t.

$$x_1 \leq 20$$

$$x_2 \leq 30$$

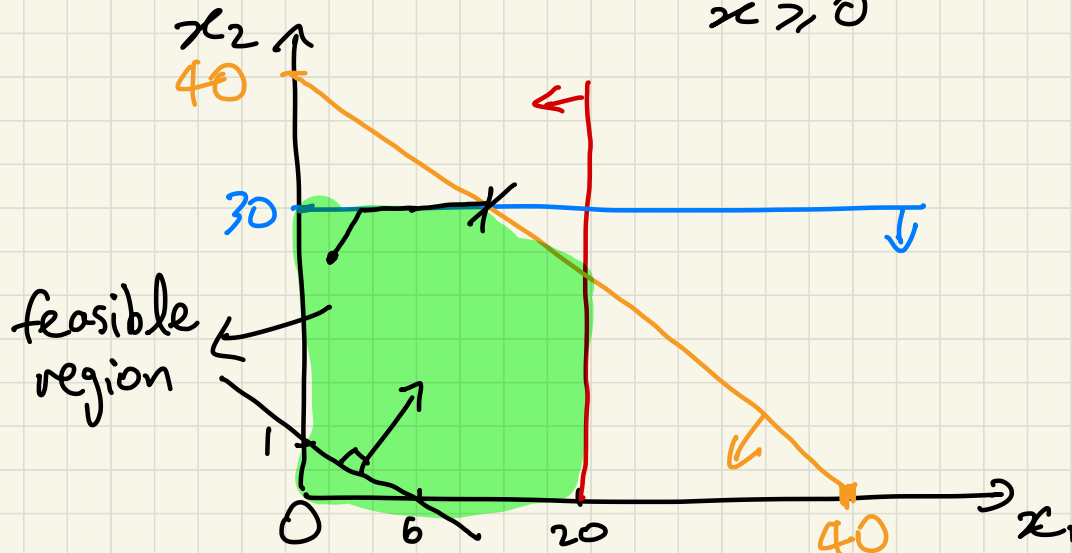
$$x_1 + x_2 \leq 40$$

$$x \geq 0$$

$$a = (1, 6)$$

$$b = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$c = \begin{pmatrix} 20 \\ 30 \\ 40 \end{pmatrix}$$



$$x_1 + 6x_2 = 6$$

$$\Rightarrow x_1 = 0, x_2 = 1$$

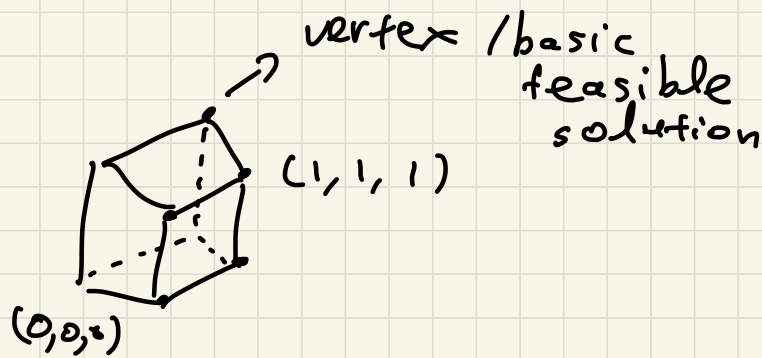
$$\Rightarrow x_2 = 0, x_1 = 6$$

$$x \in \mathbb{R}_{\geq 0}^3$$

$$x_1 \leq 1$$

$$x_2 \leq 1$$

$$x_3 \leq 1$$



$$x \in \mathbb{R}^n$$

$$B \in \mathbb{R}^{m \times n}$$

$$c \in \mathbb{R}^m$$

Polytopes / Polyhedra
Polyhedron

x is basic feasible solution

$$\text{to } Bx \leq c$$

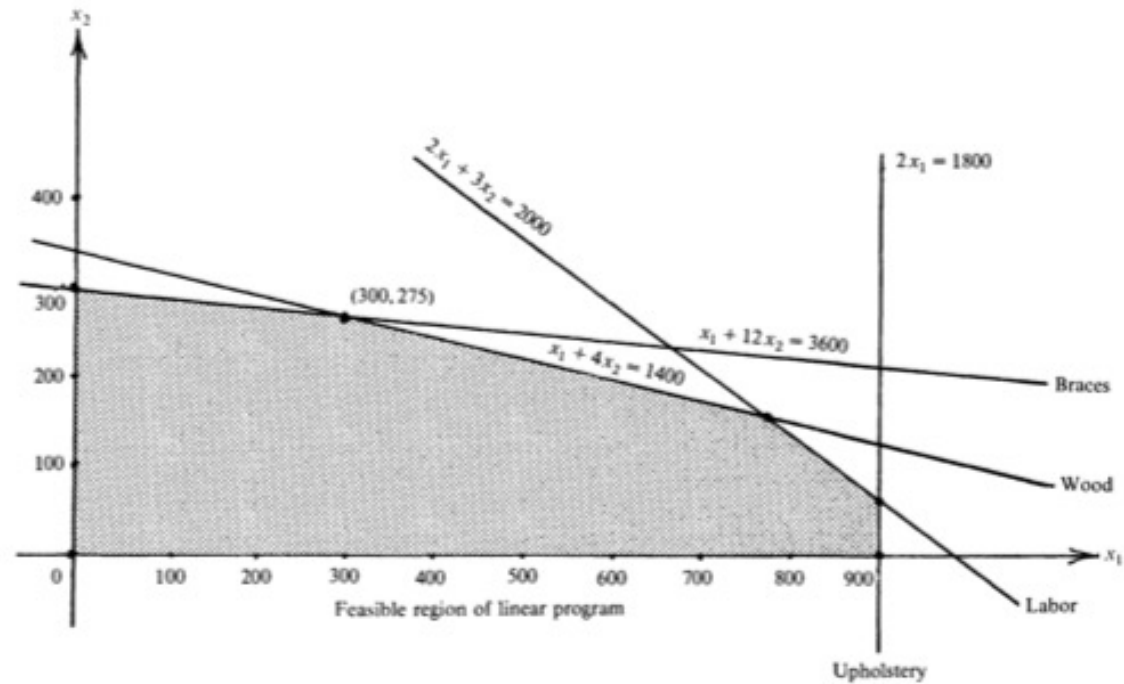
if x satisfies n constraints exactly.

$$\sum_j b_{ij} x_j = c_i \quad (\text{tight constraint})$$

Example of a Linear Program. A factory can manufacture chairs and tables. Let x_1 be the number of chairs produced and x_2 the number of tables. Chairs sell for \$40 apiece and tables for \$200 apiece. We seek to maximize our sales income, subject to limited supplies of raw materials. The inequalities below describe the production requirements and constraints on these materials:
 Maximize $z = 40x_1 + 200x_2$ subject to $x_1 \geq 0$, $x_2 \geq 0$, and

$$\begin{array}{ll} \text{Wood:} & x_1 + 4x_2 \leq 1400 \\ \text{Labor:} & 2x_1 + 3x_2 \leq 2000 \\ \text{Braces:} & x_1 + 12x_2 \leq 3600 \\ \text{Upholstery:} & 2x_1 \leq 1800 \end{array} \quad (1)$$

The model (1) is called a linear program. The Figure below shows the *feasible region* (shaded area) of points (x, x_2) that satisfy the inequalities in (1). It is straightforward to show that a linear objective function assumes its maximum value at a corner of the feasible region (assuming the feasible region is bounded).



Simplex Method

- Basic idea
 - Start with some basic feasible solution
 - Move to a better “neighboring” basic feasible solution
 - If no better neighbor exists, return current solution
- Discovered by Dantzig
- In practice: Linear time
 - Even when implemented by hand using pen and paper!
- Worst-case analysis: Exponential time
- Beyond worst-case analysis:
 - Smoothed analysis (Godel Prize, Nevanlinna Prize, Fulkerson Prize)
 - By the book analysis (STOC 2026)

Ellipsoid Method

- Basic idea
 - Reduce optimization to checking feasibility
 - If center of ellipsoid not feasible, use separating hyperplane to update ellipsoid
 - Volume of new ellipsoid decreases by a constant factor
- Worst-case: polynomial time
- In practice: way slower than Simplex
- Takeaway: Suffices to be able to efficiently determine if a solution is feasible and if not, to produce a violated constraint
- Application: Path LP for min (s,t)-cut